Cpts 317. 10/3

midterm Review.

① 4 page single-sided cheating sheets allowed.

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②. Control time.

③. items:

Write regExp from English description;

Draw FA from regExp;

Conversions/Constructions:

NFA to DFA (subset construction)

Λ-NFA to NFA (Λ-elimination).

Cartesian Product (intersection)

Complement Construction (hw).

structural induction (Induction Hypothesis formulation).

reg. language operations:

do proofs using these.

$L_1 \cup L_2$ is reg if L_1 & L_2 are reg;

$L_1 \cap L_2$ . . . . . ;

$\overline{L_1}$ . . . . . if L_1 is reg;

FA vs C-program.

1. Write RegExp.

$\Sigma = \{0, 1\}$.

(1). all strings ended with 0100.

$(0+1)^* 0100$

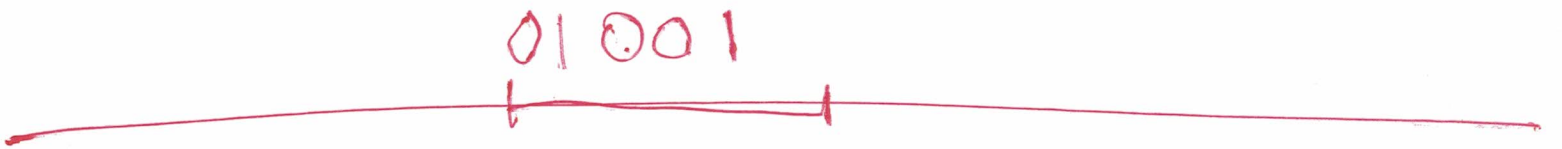
(2). all strings containing 0100 as substring

$(0+1)^* 0100 (0+1)^*$.

(3). all strings containing both 0100 and 1001 as substrings.

$$(0+1)^* 0100 (0+1)^* 1001 (0+1)^* \quad +$$

$$(0+1)^* 1001 (0+1)^* 0100 (0+1)^* \quad +$$



$$(0+1)^* 01001 (0+1)^* \quad +$$

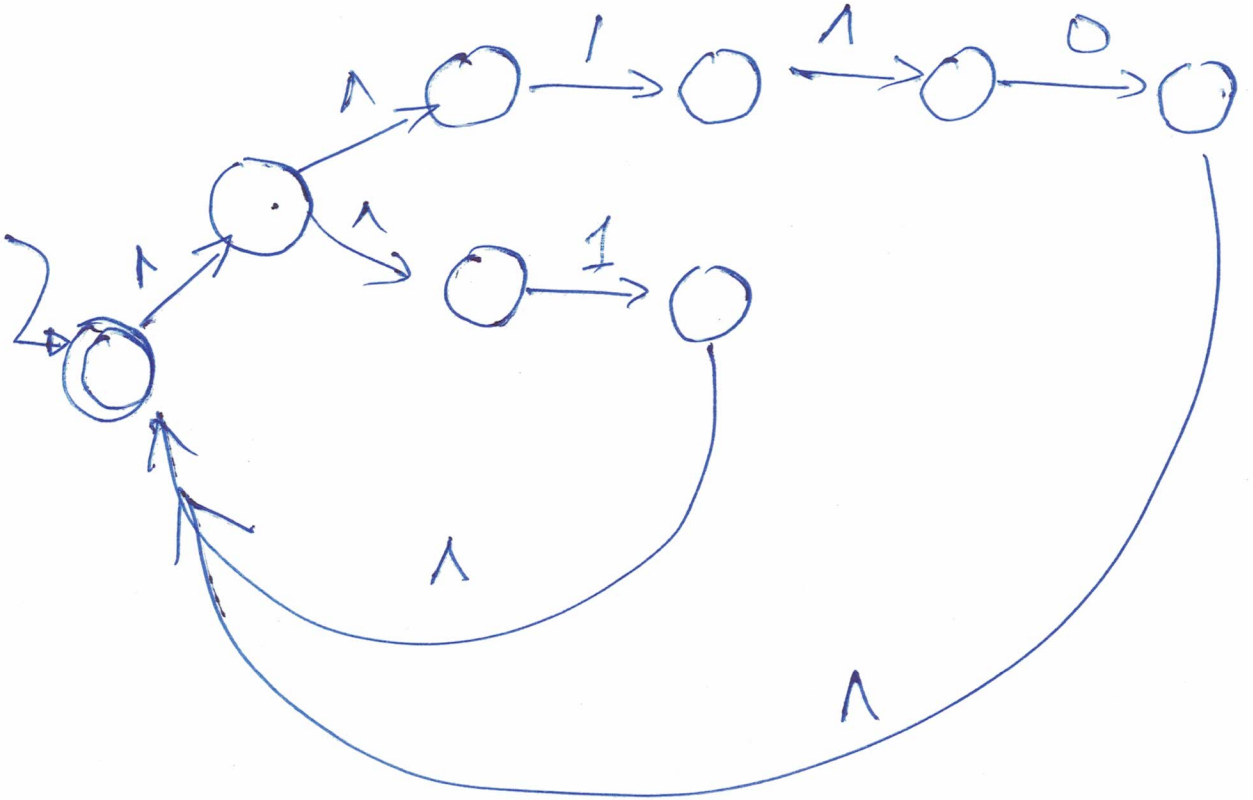
100100

$(0+1)^* 100100 (0+1)^*$

2. Draw FA from Reg Exp.

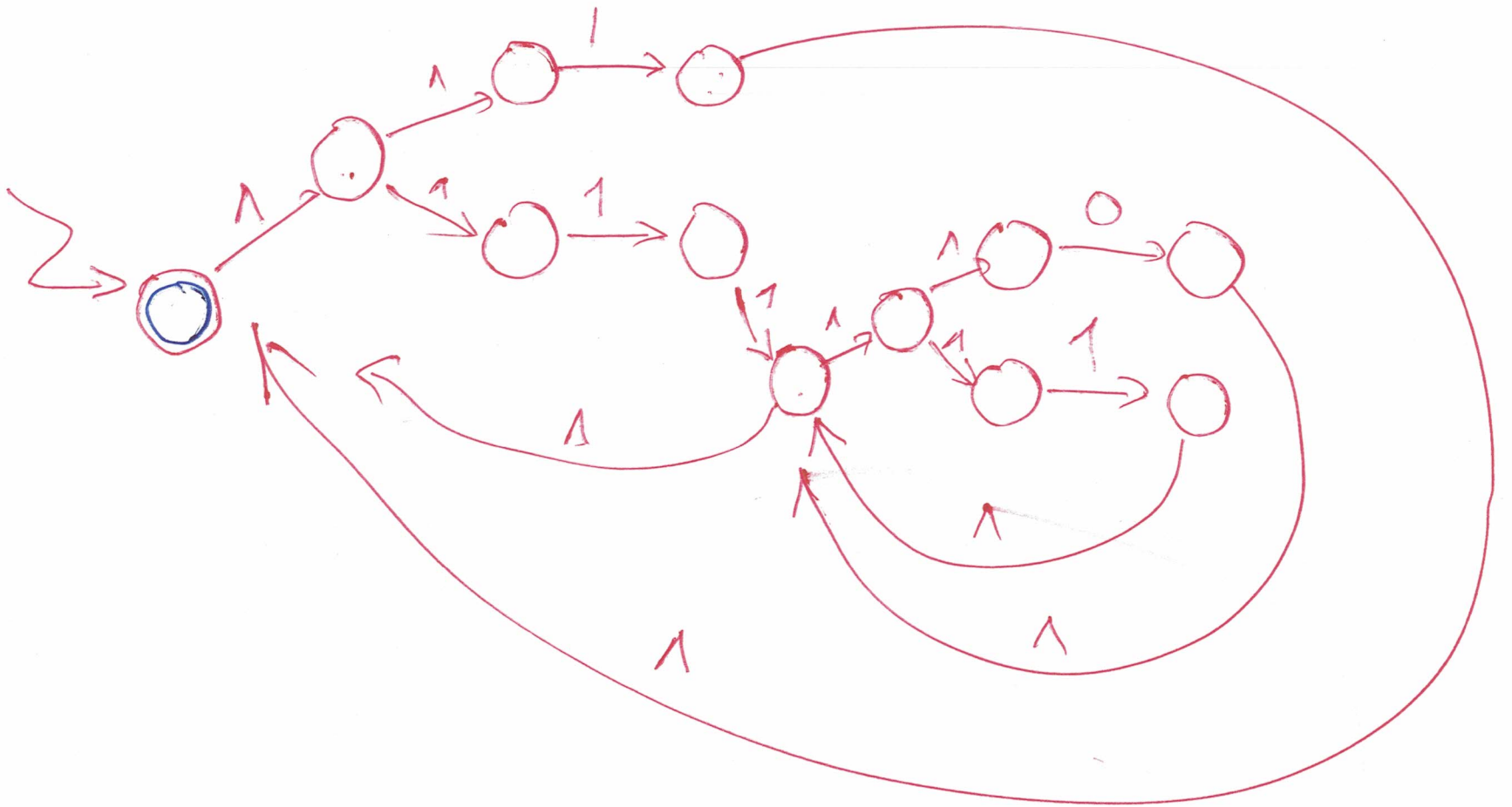
① Use Kleene Thm.

Exaple. Use Kleene Thm to draw a FA for reg Exp $(10+1)^*$:


1
1
0
1
1
1
1
1
1

use Kleene Thm to draw FA example:

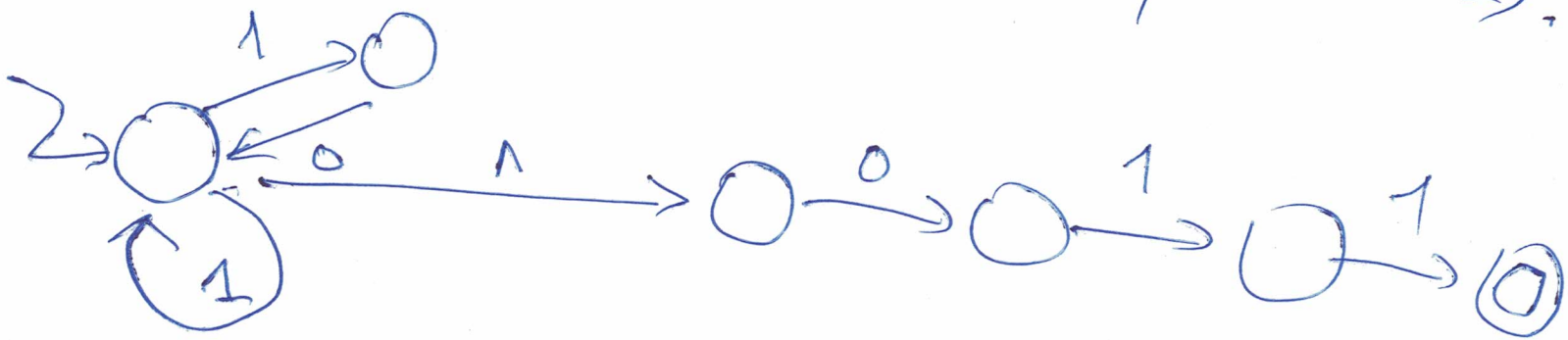
$$(1 + 1(0+1)^*)^*$$



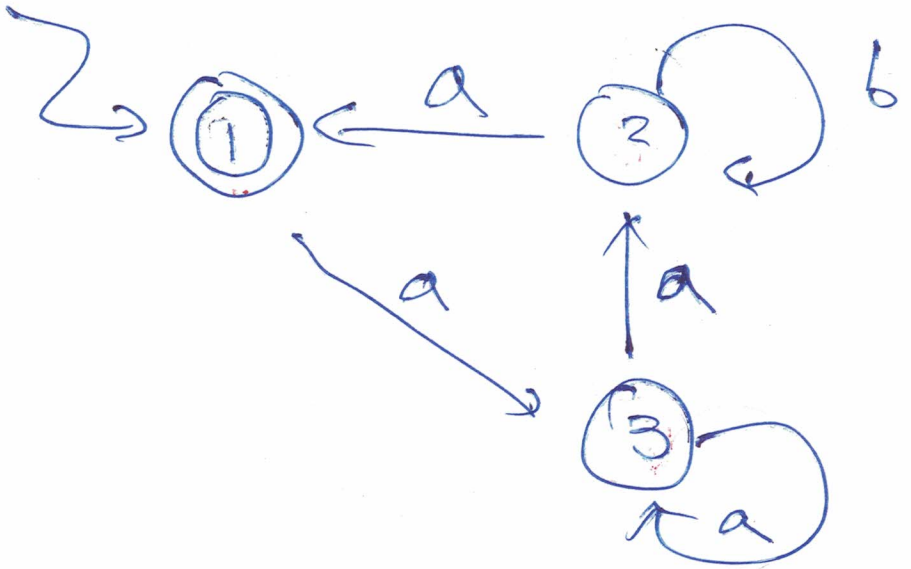
② Direct Draw.

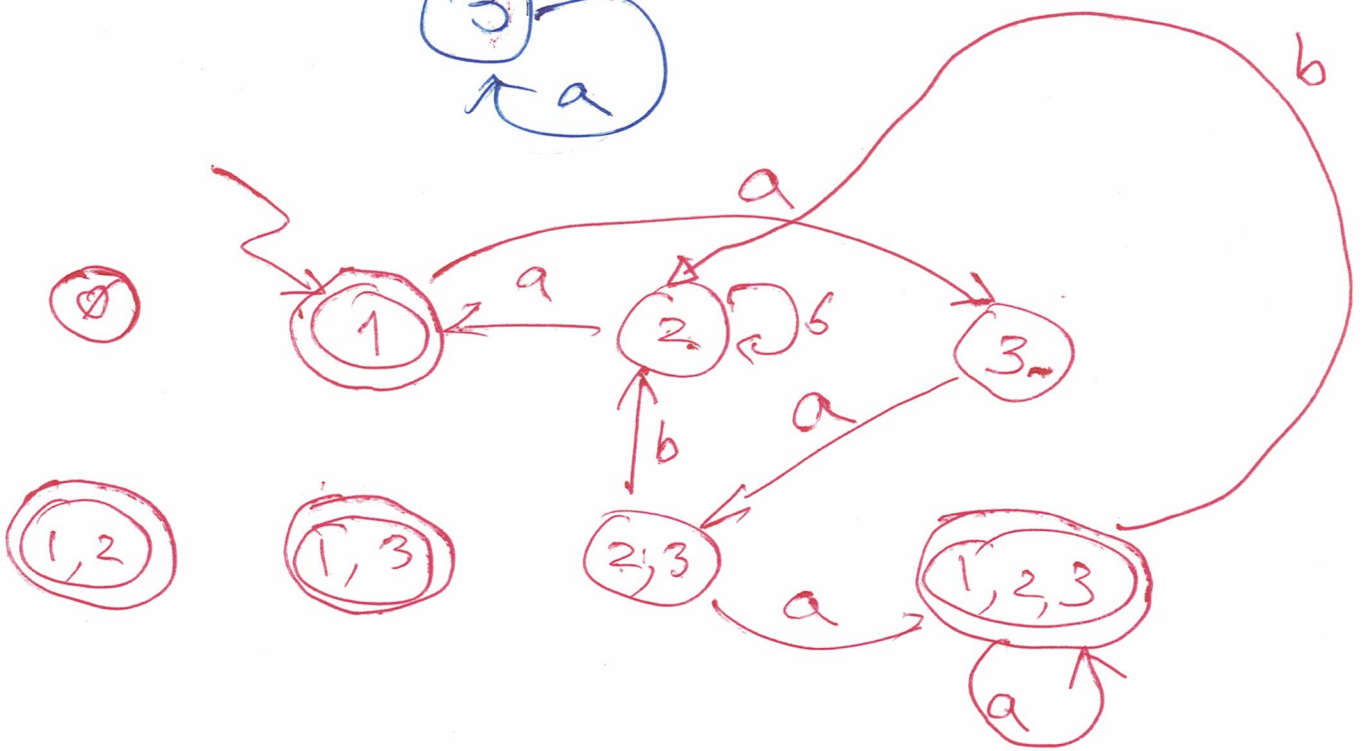
Example. Draw a FA for $(10+1)^* 011$

(FA = DFA, NFA, Λ-NFA at your choice).

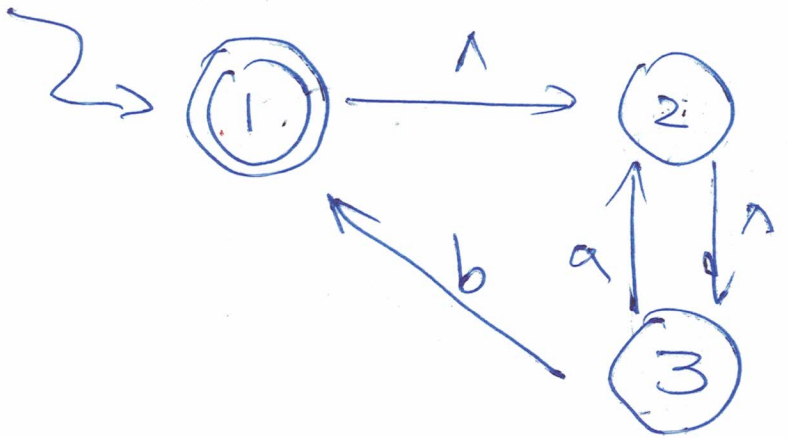


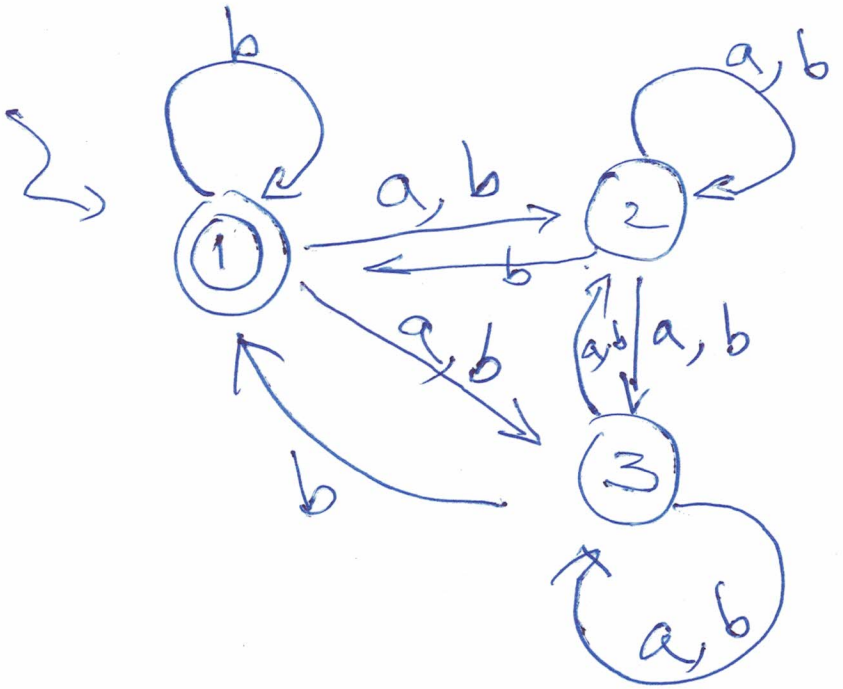
3. NFA to DFA (subset construction).





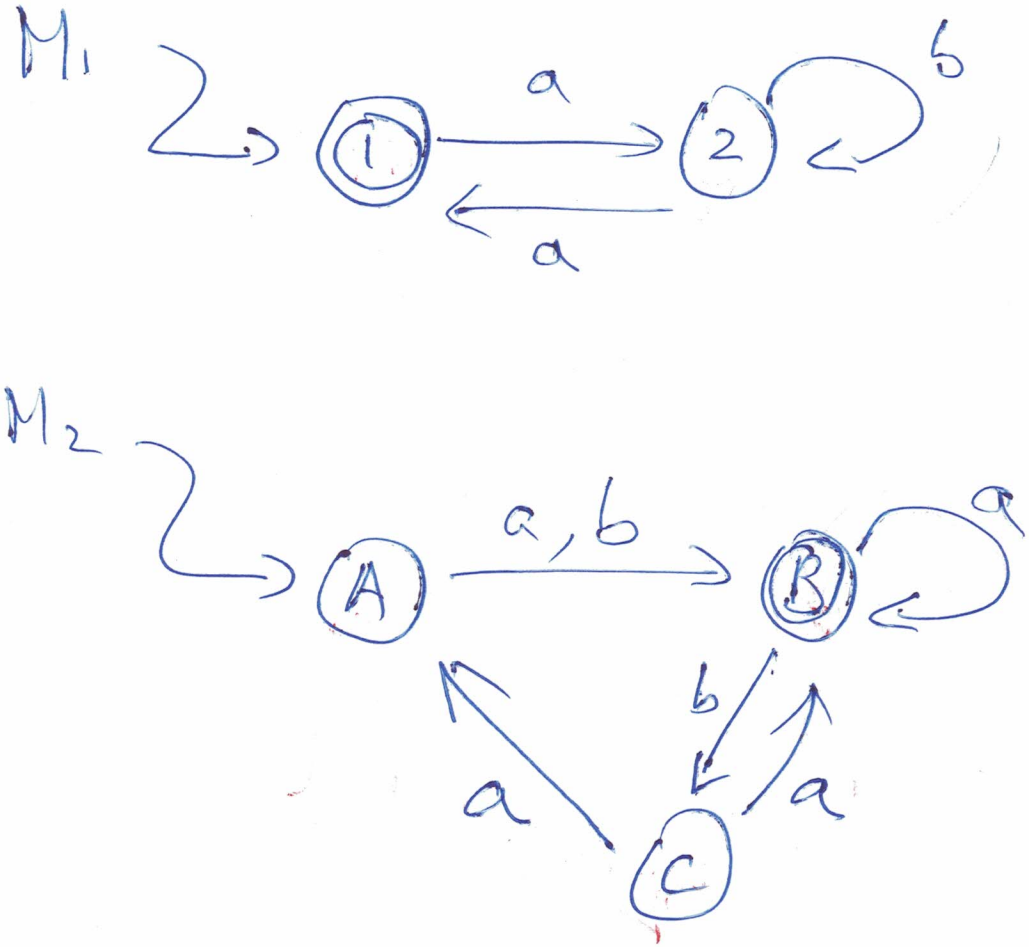
Λ-NFA to NFA:

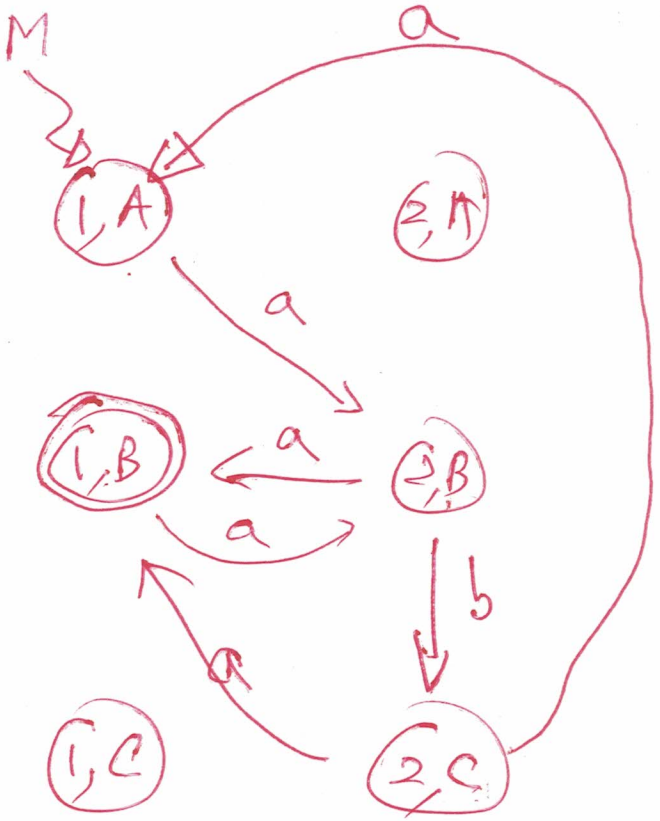




Cartesian Product.

Draw a FA M with $L(M) = L(M_1) \cap L(M_2)$.



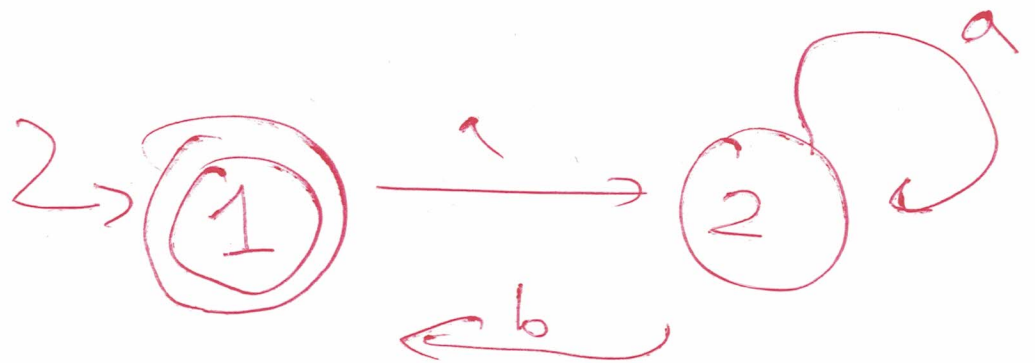


structural induction:

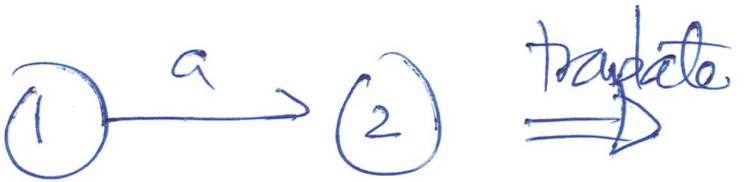
look at class notes. (also hw).

DFA & C-programs.

M



in your mind:



when x==1 & we read 'a', do: x:=2,

```
x := 1;
while(1) {                    if x==1 and end of input,
                              ~~break~~ halt & happy;
  if x == 1 & read('a') then
        x := 2 else
  if x == 2 & read('a') then
        x := 2 else
  if x == 2 & read('b') then
        x := 1;

}
```

Show: $L_1 - L_2 = \{w : w \in L_1 \text{ and } w \notin L_2\}$

is reg if L_1 & L_2 are reg.

Proof. Note that

$L_1 - L_2 = L_1 \cap \overline{L_2}$ and

reg-langs are closed under

$\cap$ and $\overline{}$. $\boxtimes$